

DATABASE MANAGEMENT SYSTEM LAB

ASSIGNMENT NO. 01 – [Name : Ridafatema Patel Rollno. 63]

Consider the given Database Schema :

```
MariaDB [RIDA]> SELECT * FROM Employee;SELECT * FROM Works;SELECT * FROM Company;SELECT * FROM Manager;
```

EmpId	Name	Street	City
101	Sahil Khan	Colony	Nasik
102	Gaurav Pagare	Jailroad	Nasik
103	Om Dhole	Dhruv Nagar	Nasik
104	Satyam Singh	Jadhav Sankul	Nasik
105	Siddhesh Chavan	Cidco	Nasik

5 rows in set (0.000 sec)

EmpName	Company	Salary
Sahil Khan	First Bank Corporation	50000
Gaurav Pagare	Small Bank Corporation	40000
Om Dhole	ThoughtWorks	60000
Satyam Singh	Whirlpool	45000
Siddhesh Chavan	Sumago	30000

5 rows in set (0.000 sec)

Company	Name	City
First Bank Corporation	Rida Patel	Nasik
Small Bank Corporation	Gayatri Dhakane	Mumbai
ThoughtWorks	Damini Pawar	Pune
Whirlpool	Prerana Mahajan	Nasik
Sumago	Saipriya Khedlekar	Pune

5 rows in set (0.000 sec)

EmpName	ManagerName
Sahil Khan	Rida Patel
Gaurav Pagare	Gayatri Dhakane
Om Dhole	Damini Pawar
Satyam Singh	Prerana Mahajan
Siddhesh Chavan	Saipriya Khedlekar

Write SQL queries for the following :-

1. Find the names of all employees who work for First Bank Corporation.

```
MariaDB [RIDA]> SELECT EmpName FROM Works WHERE Company = 'First Bank Corporation';
```

EmpName
Sahil Khan

2. Find the names and cities of residence of all employees who work for First Bank Corporation

```
MariaDB [RIDA]> SELECT e.Name, e.City FROM Employee e JOIN Works w ON e.Name = w.EmpName WHERE w.Company = 'First Bank Corporation';
```

Name	City
Sahil Khan	Nasik

3. Find the names, street addresses, and cities of residence of all employees who work for First

```
MariaDB [RIDA]> SELECT e.EmpId, e.Name, e.Street, e.City FROM Employee e JOIN Works w ON e.Name=w.EmpName WHERE w.Company = 'First Bank Corporation' AND w.Salary >10000;
```

EmpId	Name	Street	City
101	Sahil Khan	Colony	Nasik

4. Find all employees in the database who live in the same cities as the companies for which they work.

```
MariaDB [RIDA]> SELECT e.Name FROM Employee e JOIN Works w ON e.Name = w.EmpName JOIN Company c ON w.Company = c.Company WHERE e.City =c.City;
```

Name
Sahil Khan
Satyam Singh

5. Find all employees in the database who live in the same cities and on the same streets asdo

```
MariaDB [RIDA]> SELECT e.Name FROM Employee e JOIN Manager m ON e.Name = m.EmpName JOIN Employee mgr ON m.ManagerName = mgr.Name WHERE e.City = mgr.City AND e.Street = mgr.Street;
```

Empty set (0.005 sec)

6. Find all employees in the database who do not work for First Bank Corporation.

```
MariaDB [RIDA]> SELECT EmpName FROM Works WHERE Company <> 'First Bank Corporation';
```

EmpName
Gaurav Pagare
Om Dhole
Satyam Singh
Siddhesh Chavan

7. Find all employees in the database who earn more than each employee of Small Bank Corporation.

```
MariaDB [RIDA]> SELECT EMPName FROM Works WHERE Salary > ALL(SELECT Salary FROM Works WHERE Company = 'Small Bank Corporation');
```

EMPName
Sahil Khan
Om Dhole
Satyam Singh

8. Assume that the companies may be located in several cities. Find all companies located in every city in which Small Bank Corporation is located.

```
MariaDB [RIDA]> SELECT Company FROM Company WHERE City IN(SELECT City FROM Company WHERE Company = 'Small Bank Corporation');
```

Company
Small Bank Corporation

9. Find all employees who earn more than the average salary of all employees of their company.

```
MariaDB [RIDA]> SELECT w.EmpName FROM Works w WHERE w.Salary >(SELECT AVG(w2.Salary)FROM Works w2 WHERE w.Company = w2.Company);
```

Empty set (0.001 sec)

10. Find the company that has the most employees.

```
MariaDB [RIDA]> SELECT Company FROM Works GROUP BY Company ORDER BY COUNT(EmpName) DESC LIMIT 1;
```

Company
First Bank Corporation

11. Find the company that has the smallest payroll.

```
MariaDB [RIDA]> SELECT Company FROM Works GROUP BY Company ORDER BY SUM(Salary) LIMIT 1;
```

```
+-----+  
| Company |  
+-----+  
| Sumago  |  
+-----+
```

12. Find those companies whose employees earn a higher salary, on average, than the average salary at First Bank Corporation.

```
MariaDB [RIDA]> SELECT Company FROM Works GROUP BY Company HAVING AVG(Salary)>(SELECT AVG(Salary) FROM Works WHERE Company = 'First Bank Corporation');
```

```
+-----+  
| Company |  
+-----+  
| ThoughtWorks |  
+-----+
```